BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

LESSON 292 · CSS and JavaScript

Name: _____ Class: _____ Date: _____

Time: _____ Teacher: _____

Work through the pages in order. Answer in BLACK PEN first. Check in GREEN PEN only after the Exit Ticket.

Do Now · Recall previous learning

Q1. What does the HTML id attribute identify?

Q2. What does a CSS selector do?

Q3. What does a button click trigger in JavaScript?

Q4. Binary search in [2,4,6,8,10,12,14] for 12: show midpoint indices.

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I Do · Read new knowledge

Key vocabulary

CSS — Style sheets controlling presentation.

selector — Targets HTML/DOM nodes.

DOM — Browser's document object model.

event handler — Code invoked by event.

textContent — Text stored in a DOM node.

New learning

Understand — CSS controls presentation using selectors and declarations; `#status` targets one element by id.

Trace — JavaScript can react to browser events and modify the DOM. The handler does not execute until a click event occurs.

Evaluate — The document's structure remains HTML, styles remain CSS, and interactive behaviour is JavaScript. Validate incoming data on the server even if scripts check it first.

Worked example

Trace click count initial 0 when handler increments by one on each click. Two clicks occur.

1. Predict first — Start with count=0; first click handler adds one making count=1.

2. Show method — Second click adds one again giving count=2; no change happens before an event.

We Do · Practise with guidance

A page contains p id='msg' showing 'Off'. Click handler changes textContent to 'On'. Explain DOM result.

```
document.getElementById('msg').textContent='On';
```

Predict first

getElementById selects the existing paragraph whose id is 'msg'.

Show method

Assigning textContent replaces displayed text with 'On'; it is safer for text than injecting untrusted HTML.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

You Do · Work independently

BLACK PEN: Write your own answers. Do not look at the Review & Improve pages yet.

Question 1

In the code, what does CSS #status apply and what is initial text? [4]

Question 2

Write JavaScript so a click on button id='go' changes p id='status' to 'Complete'. [6]

Question 3

Compare changing text with textContent and inserting untrusted input via innerHTML. [6]

Exit Ticket · Answer before checking

JavaScript is needed to define every HTML heading and paragraph.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Review & Improve · GREEN PEN

Now change to GREEN PEN. Find each original question. Tick correct answers; read the model and rewrite mistakes.

Page 1 onwards · Do Now

Q1. A specific DOM element.

Q2. Targets elements to style.

Q3. A registered event handler.

Q4. mid3→8; low4 high6 mid5→12, found indexes 3,5.

You Do pages · Independent questions

Q1. Select element with id status [2]; style color blue/font bold [1]; initial text Waiting [1].

Q2. getElementById('go') [1]; addEventListener('click',...) [2]; getElementById('status') [1]; .textContent='Complete' [2].

Q3. textContent treats input as text [2]; innerHTML parses markup [2]; malicious input may inject scripts/content [1]; use safe escaping/DOM APIs [1].

Exit Ticket · Model answer

False. HTML defines structure; JS adds behaviour.

Try the question again

Choose one question you got wrong. All correct? Choose the hardest.

1. Copy the whole original question:

2. Write a complete new answer in GREEN PEN. Show working if needed: